A MONOMORPHIC FIELD-VALUED POINT ABSORBS EVERY FIELD-VALUED POINT OF THE SAME EQUIVALENCE CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 03DP: sources/stacks-chow/tags.txt, source lines 4428–4428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $p : \mathrm{Spec}(K) \to X$ and $q : \mathrm{Spec}(L) \to X$ be morphisms where K and L are fields. Assume p and q determine the same point of $|X|$ and p is a monomorphism. Then q factors uniquely through p.*

Proof. Since p and q define the same point of $|X|$, we see that the scheme

$$Y = \mathrm{Spec}(K) \times_{p,X,q} \mathrm{Spec}(L)$$

is nonempty. Since the base change of a monomorphism is a monomorphism this means that the projection morphism $Y \to \mathrm{Spec}(L)$ is a monomorphism. Hence $Y = \mathrm{Spec}(L)$, see Schemes, Lemma 03DP. We conclude that q factors through p. Uniqueness comes from the fact that p is a monomorphism. □

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